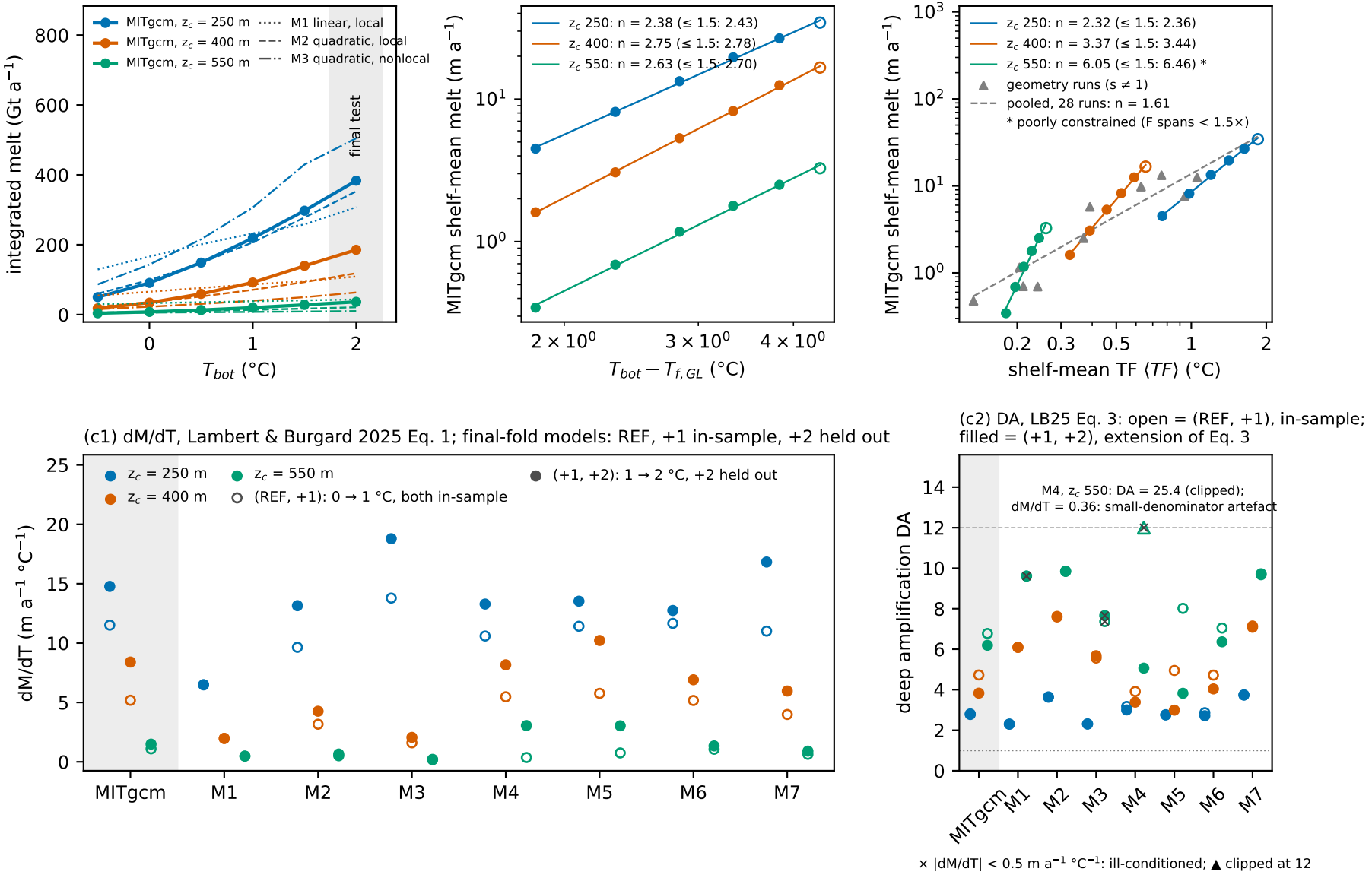
(a) integrated melt, held-out predictions
MITgcm, z_c = 250 m
MITgcm, z_c = 400 m
MITgcm, z_c = 550 m
M1 linear, local
M2 quadratic, local
M3 quadratic, nonlocal
final test
integrated melt (Gt a^{-1})
T_{bot} (°C)
(b1) $\langle m \rangle = kF^n$ (POST HOC vs Exp. 2)
z_c 250: n = 2.38 (≤ 1.5: 2.43)
z_c 400: n = 2.75 (≤ 1.5: 2.78)
z_c 550: n = 2.63 (≤ 1.5: 2.70)
MITgcm shelf-mean melt (m a^{-1})
$T_{bot} - T_{f,GL}$ (°C)
(b2) $\langle m \rangle = kF^n$ (POST HOC vs Exp. 2)
z_c 250: n = 2.32 (≤ 1.5: 2.36)
z_c 400: n = 3.37 (≤ 1.5: 3.44)
z_c 550: n = 6.05 (≤ 1.5: 6.46) *
geometry runs (s ≠ 1)
pooled, 28 runs: n = 1.61
* poorly constrained (F spans < 1.5×)
MITgcm shelf-mean melt (m a^{-1})
shelf-mean TF $\langle TF \rangle$ (°C)
(c1) dM/dT, Lambert & Burgard 2025 Eq. 1; final-fold models: REF, +1 in-sample, +2 held out
z_c = 250 m
z_c = 400 m
z_c = 550 m
(REF, +1): 0 → 1 °C, both in-sample
(+1, +2): 1 → 2 °C, +2 held out
dM/dT (m a^{-1} °C^{-1})
MITgcm
M1
M2
M3
M4
M5
M6
M7
(c2) DA, LB25 Eq. 3: open = (REF, +1), in-sample; filled = (+1, +2), extension of Eq. 3
M4, z_c 550: DA = 25.4 (clipped);
dM/dT = 0.36: small-denominator artefact
deep amplification DA
× |dM/dT| < 0.5 m a^{-1} °C^{-1}: ill-conditioned; ▲ clipped at 12